BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

BIOLOGY

Past Paper 2024 • Paper Code: 8463

Retyped faithfully from the original board paper — same questions, no changes.

Biology - Paper: II Objective (ii)

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Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8463

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Q1. In a nucleotide, a nitrogenous base is attached to carbon number:

(A) 5
(B) 4
(C) 3
(D) 1

Q2. A set of three nucleotides on mRNA specifying a particular amino acid is called:

(A) Code
(B) Genetic code
(C) Codon
(D) Anticodon

Q3. Brothers having same parents are not similar due to:

(A) Mitosis
(B) Synopsis
(C) Apoptosis
(D) Crossing over

Q4. On looking at a pure white cat, a student of genetics said, "Alas, the cat is deaf!". Guess the genotype of cat.

(A) W/W
(B) W/w
(C) w/w
(D) Both A & B

Q5. Indicate the salt tolerant plant among the given:

(A) Acacia
(B) Mango
(C) Arabidopsis
(D) Shisham

Q6. In which disease a patient lacks a gene coding for trans-membrane carrier of chloride ions?

(A) SCID
(B) Cystic fibrosis
(C) Cancer
(D) Hodgkin's lymphoma

Q7. Which protein is similar in all aerobic organisms?

(A) Cytochrome b
(B) Haemoglobin
(C) Cytochrome c
(D) Both A & C

Q8. The term niche was coined by:

(A) Joseph Grinnell
(B) Charles Elton
(C) Tansley
(D) Hult

Q9. Select the inappropriate matching.

(A) Khanpur ↔ Cholistan
(B) Yazman ↔ Thar
(C) Mianwali ↔ Thal
(D) Gilgit ↔ Grassland

Q10. Homo sapiens have been on the earth for about:

(A) 10,000 years
(B) 20,000 years
(C) 40,000 years
(D) 60,000 years

Q11. Which of the given retards cell elongation in plants?

(A) Red light
(B) Blue light
(C) Ultraviolet rays
(D) Both A & C

Q12. An individual produced by parthenogenesis is:

(A) Male
(B) Female
(C) Male or female
(D) Hermaphrodite

Q13. Transient alteration in brain due to excessive rapid electrical discharges in the grey matter are diagnosed as:

(A) Epilepsy
(B) Parkinsonism
(C) Alzheimer's disease
(D) Goiter

Q14. Hormone that stimulates conversion of glucose into lipids and proteins is:

(A) Cortisol
(B) Insulin
(C) Aldosterone
(D) Glucagon

Q15. What is not true about sclerenchyma?

(A) Lignin
(B) Branched pits
(C) Protoplast
(D) Thick cell wall

Q16. Nastic movements are due to balance or ratio between:

(A) Abscisic acid and gibberellin
(B) Abscisic acid and cytokinin
(C) Abscisic acid and auxins
(D) Both A & C

Q17. Production of heat or high temperature during fever is called:

(A) Pyrogen
(B) Pyrexia
(C) Antitoxin
(D) Both A & C

Biology - Paper: II Subjective

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Time Allowed: 2:40 Hours | **Maximum Marks:** 68

> Note: Note:- Section B is compulsory.

Q2.. (Marks 16) Write short answers to any EIGHT parts. (8 x 2 = 16)

(i) How do fresh water organisms maintain osmoregulation?

(ii) Human nephrons have association of three types of capillary beds. Give their names and location in the kidney.

(iii) Why leaves are said to be excretophores?

(iv) What do you know about skeletal deformities because of genetic causes? Discuss any two of them.

(v) What kind of cells are responsible for bone formation?

(vi) Name unpaired bones of cranium.

(vii) How would you differentiate between oviparous and viviparous condition?

(viii) What is the importance of pollen tube in spermatophytes?

(ix) What is profundal zone? What is its source of nutrition?

(x) Give two adaptations in plants and animals for terrestrial ecosystem.

(xi) How chlorine is responsible for ozone depletion?

(xii) What is acid rain? Write its any two effects.

Q3.. (Marks 16) Write short answers to any EIGHT parts. (8 x 2 = 16)

(i) Define Biological Rhythms. Write names of its types.

(ii) Write functions of cerebellum.

(iii) How would you define innate behaviour? Give example.

(iv) What do you remember about law of independent assortment?

(v) Why blood group O is called universal donor?

(vi) What do you understand by sex limited trait? Give example.

(vii) Write names of things which are required to produce recombinant DNA.

(viii) What is restriction fragment length polymorphism? How is it detected?

(ix) How would you define Transgenic Organisms?

(x) How does a predator affect prey and vice versa?

(xi) What does happen in denitrification?

(xii) How would you compare autecology and synecology?

Q4.. (Marks 12) Write short answers to any SIX parts. (6 x 2 = 12)

(i) Define lateral meristem. Give example.

(ii) How coelom is formed in chick embryo?

(iii) What is nucleosome?

(iv) Differentiate between leading strand and lagging strand of DNA.

(v) How does phenylketonuria affect body?

(vi) Define cell cycle. Give its sketch also.

(vii) What is malignant tumour?

(viii) Comparative embryology is an evidence of evolution. Justify.

(ix) Define Hardy-Weinberg Theorem.

> Note: SECTION-C Note:- Attempt any THREE questions. Each question carries EIGHT (4+4=8) marks. (8x3=24)

Q5..

(a) How are excretory products made concentrated in the human kidney?

(b) Write down four differences between mitosis and meiosis.

Q6..

(a) How are broken bones repaired? Explain.

(b) What is succession? Explain it with an example of xerosere.

Q7..

(a) Discuss the main events involved in initiation of nerve impulse.

(b) Write down any four factors which may affect gene frequency.

Q8..

(a) Describe the role of phytochrome in photoperiodism.

(b) In monohybrid cross, we get 3:1 phenotype and 1:2:1 genotype ratio. Prove it with one example.

Q9..

(a) Write a note on embryonic induction.

(b) What is polymerase chain reaction? How does it work? Give its applications also.